

1604000 - Wastewater	
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Spec Item	Title	Description
1	General Information	
1.1	Outline of Services	The intent of this Annex is to specify the requirements to provide Wastewater utilities. Included are services such as maintaining, and repairing the wastewater treatment plant and collection system, and other associated equipment and systems. Service Order response for Wastewater is included in 1601000 Utility Management.

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2	Management and Administration	
2.1	Definitions and Acronyms	Definitions and Acronyms are listed in J-1604000-01.
2.2	Personnel	The Contractor shall provide personnel with the qualifications, technical knowledge, experience and skills required to efficiently operate, maintain and repair the Wastewater Treatment Plant and Collection System.
2.2.1	Certification, Training and Licensing	The Contractor shall provide certified personnel with the qualifications, technical knowledge, experience and skills required to efficiently maintain and repair the wastewater treatment plant and collection system. The Contractor shall submit proof of all certifications, training, and licensing requirements per Section F.
2.3	Special Requirements	
2.3.1	Wastewater Equipment	Wastewater instrumentation equipment shall be certified and calibrated annually by an approved certifying office. Wastewater equipment shall be calibrated and certified in accordance with UNI EN ISO/IEC 17025. Calibration certificates shall be accredited by ACCREDIA. Calibration reports shall be developed and provided in accordance with ISO 9001, per Section F.
2.4	References and Technical Documents	References and Technical Documents are listed in J-1604000-02.

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Spec Item	Title	Performance Objective	Related Information	Performance Standard
3	Recurring Work	The Contractor shall efficiently maintain wastewater treatment plant equipment and collection system to safely collect, treat and dispose of wastewater to ensure all sewage requirements are continually met.	The Contractor shall maintain wastewater collection and disposal systems in accordance with ASME B40.100-2005, FGS Italy C3.3.2.7, and UFC 3-230-02, Ch. 8 (8-3), FGS Italy, equipment manufacturer's instructions, industry standards, and references and technical documents listed in J-1604000-02. The current Wastewater Treatment Plant and Collection System Description and Inventory and Minimum Frequency of Services is provided in J-1604000-03. The Contractor shall develop and submit a Critical Spare Parts List per Section F. The Contractor shall maintain on hand the critical spare parts required to keep the entire system in continuous service to meet wastewater requirements. The Historical Critical Spare Parts List is provided in J-1604000-04. The Contractor is fully responsible for ensuring the critical spare parts list is updated and expanded as required and parts are available at all times. The Government will inspect maintained spare parts quarterly to ensure adequate storing and ordering levels.	Sewage is continuously and safely collected, treated and disposed. No spills are the result of Contractor negligence. Critical spare parts are on-hand and available at all times. All work performed in compliance with regulations and standards as specified.
3.1	Preventive Maintenance (PM) Program	The Contractor shall develop and implement a PM program for wastewater collection systems and wastewater treatment to ensure proper operation, to minimize breakdowns, to maximize useful life and meet all applicable discharge permit requirements.	The Contractor shall develop and submit a PM program per Section F. The Contractor is fully responsible for and shall perform any repairs, including replacement, discovered during scheduled maintenance work up to a total of \$3,000 per occurrence in direct material and labor cost under FFP portion of the contract. Repair work performed under maintenance are not considered a Service Order. Notification of repair work exceeding the incidental repairs limit shall be submitted to the KO	Maintenance is accomplished in accordance with the Contractor's maintenance program plan and work schedule. PM is performed in accordance with manufacturers' recommended procedures, OEM standards, ASME, WEF, and NASSCO.

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			within two hours of identification. Service Orders or Non-Recurring Work Task Orders work may be issued for repairs exceeding the incidental repairs limit. The Contractor shall not use breakdown maintenance as part of the PM program. The PM program for wastewater systems shall provide economical approach manufacturer's recommended procedures, OEM standards, NASSCO standards, WEF standards, ASME standards, and maintenance required to satisfy equipment warranties and keep wastewater systems in normal working condition and in compliance with the discharge permit requirements. Excessive or repeated system or equipment breakdowns or deficiencies may indicate the need to adjust or modify the Contractor's PM program. These changes will be made at no additional cost to the Government. If the KO determines that the objective and standards are not met, the Contractor shall adjust or modify the Contractor's PM program at no additional cost to the Government. The Contractor shall submit a Monthly PM Work Schedule per Section F.	Unscheduled interruptions are minimized such that collection demand and effluent quality requirements are met, 99.9 percent of the time annually.
3.1.1	WWTP Equipment Maintenance	The Contractor shall perform WWTP equipment maintenance to ensure proper operation, to minimize breakdowns, and to maximize useful life.	The Contractor shall maintain WWTP equipment and systems in accordance with manufacturers' recommended procedures, OEM standards.	Maintenance is performed in accordance with Contractor's PM program and work schedule.

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3.1.1.1	Mixer and Submersible Pumps	The Contractor shall maintain mixer and submersible pumps to ensure proper operation, to minimize breakdowns, and to maximize useful life.	The Contractor shall maintain equipment in accordance with manufacturers' recommended procedures, OEM standards. Maintenance includes lubrication of bearings, oil changes and part replacements as required.	PM is accomplished per the Contractor's program and work schedule.
3.1.2	Lift Stations	The Contractor shall perform lift station maintenance to ensure proper operation, to minimize breakdowns, and to maximize useful life.	The Contractor shall repair all concrete spalling, perform repair, and install a complete epoxy coating of the lift station. The lift station shall be fully restored and repaired to meet the latest ASME and NASSCO standards. The Contractor shall submit Pre-repair and Post-repair Reports on Lift stations in accordance with NASSCO standards per Section F.	All work is performed in accordance with the Contractor's maintenance program plan and work schedule. Lift station maintenance program meets the latest ASME and NASSCO standards.
3.1.3	Wastewater Treatment Plant and Lift Station Equipment Calibration	The Contractor shall provide calibration services for wastewater treatment plant and lift station equipment to ensure proper operation, to minimize breakdowns, and to maximize useful life.	The Contractor shall be responsible for ensuring all wastewater treatment plant and lift station tools and equipment and components of the instrumentation system are calibrated annually. The Contractor shall submit calibration certificates and related documentation per Section F. All calibrations shall be performed per AWWA, ANSI, and NFPA Standards, OMSI, equipment manufacturer's recommendations. Replacement of instrumentation systems equipment shall be with the same type, make, and model of the equipment being replaced.	All work is performed in accordance with the Contractor's maintenance program plan and work schedule. Services meet the latest ASME, NASSCO standards, manufacturer's recommended procedures and OEM standards.
3.1.4	Lift Station Submersible Pumps	The Contractor shall maintain sewage submersible pumps to ensure proper operation, to minimize breakdowns, and to maximize useful life.	Submersible pump maintenance shall be performed in accordance with applicable federal and local statutes and regulations, and with DoD policies, instructions and guidance.	All work is performed in accordance with the Contractor's maintenance program plan and work schedule. Submersible pump maintenance program meets manufacturer's recommended procedures and OEM standards.

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3.2	Other Recurring Services	The Contractor shall perform other recurring services for sewer systems and equipment to ensure they are fully functional and operational.	The Contractor shall perform other recurring work services on wastewater systems and equipment.	Systems and equipment are in an operable condition and function properly in accordance with specified standards. Other recurring services are completed in a timely manner.
3.2.1	Lift Stations Services	The Contractor shall clean and pump lift stations to ensure proper operations and prepare lift stations for inspection and maintenance.	The Contractor shall perform lift station services at least twice per year. The Contractor shall submit a Schedule of Lift station Services per Section F. The Contractor shall spray down lift station surfaces with a mixture of water and detergent under high pressure prior to any personnel entering the lift station. The Contractor shall prevent spillage during operations. Any spillage due to Contractor operations shall be cleaned and treated at the Contractor's expense. Any spillage of waste materials shall be recorded and reported to the KO within one hour and reported immediately to the PWD Utilities Office. Spills on or off base shall be handled in accordance with the installation Spill Containment Plan.	Services are completed per the Contractor's schedule. No spills as the result of Contractor operations. Lift stations are in normal working condition.
3.2.1.1	Lift Stations Pumping	The Contractor shall pump lift stations to ensure liquids, waste, and debris is removed.	The Contractor shall perform lift station pumping in accordance with federal, state, and local regulations. The Contractor shall use the appropriate equipment to completely pump the lift stations in a timely and efficient manner with little to no disruption to base operations. The Contractor shall provide a Lift station Volume Report of sanitary sewage pumped from lift stations per Section F.	Lift stations are completely pumped and free of all liquids, waste, and debris.

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3.2.1.2	Lift Stations Cleaning	The Contractor shall clean lift stations to ensure all waste and debris is fully removed from lift station surface and equipment.	The Contractor shall check with the KO or designated Government representative before cleaning equipment in a lift station with a power washer. The Contractor shall remove all run-off liquid from the cleaning of the lift station. The Contractor shall prepare and submit a Lift station Cleaning Report per Section F.	All surfaces and equipment are free of waste and debris.
3.2.2	WWTP Equipment Sewage Waste Collection, Removal, and Disposal	The Contractor shall provide WWTP sewage waste collection, removal, and disposal services to ensure proper WWTP operation, to minimize breakdowns, and to maximize useful life.	The Contractor shall collect, remove, transport, and dispose of all solid and liquid sewage waste. The Contractor shall dispose of all sewage waste in accordance with current environmental regulations. The Contractor shall submit a waste manifest for each sewage waste collection, removal and disposal, per Section F.	Waste is disposed at authorized locations in accordance with current environmental regulations. Services are performed in accordance with the Contractor's work schedule.
3.2.2.1	Bar Screens, Screw Filters, Grit Vortex	The Contractor shall provide bar screens, screw filters, grit vortex sewage waste collection, removal, and disposal services to ensure proper WWTP operation, to minimize breakdowns, and to maximize useful life.	The Contractor shall perform lab testing, collection, loading, transport, and disposal of waste. The Contractor shall dispose of waste at authorized locations in accordance with current regulations.	Waste is disposed at authorized locations in accordance with current environmental regulations.
3.2.2.2	Pre-Treatment Channel	The Contractor shall provide pre-treatment channel sewage waste collection, emptying bins, removal, and disposal services to ensure proper WWTP operation, to minimize breakdowns, and to maximize useful life.	The Contractor shall clean the pre-treatment channel, collect, remove, transport, and dispose of all waste and debris produced. The Contractor shall provide the appropriate sized bags for each bin that is emptied. The Contractor shall dispose of waste at authorized locations in accordance with current environmental regulations.	Waste is disposed at authorized locations in accordance with current environmental regulations.
3.2.2.3	Equalization Basin and Pit Pumps	The Contractor shall provide equalization basin and pit pump, removal, and disposal	The Contractor shall collect, remove, transport, and dispose of all sewage sludge.	Waste is disposed at authorized locations in accordance with

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		services to ensure proper WWTP operation, to minimize breakdowns, and to maximize useful life.	The Contractor shall dispose of waste at authorized locations in accordance with current environmental regulations.	current environmental regulations.
3.2.3	Drying Bed Sludge Removal and Maintenance	The Contractor shall provide dry-bed sludge removal services to ensure sludge is properly disposed.	The Contractor shall perform sewage treatment, sludge removal, and drying bed maintenance. The After cleaning the drying beds, the Contractor shall supply new sand as required to ensure full bed operations. The Contractor shall remove sludge from the dry sludge beds and haul and dispose of dry sludge from the wastewater treatment plants. The Contractor shall dispose of the sludge in accordance with OEBGD guidance for hazardous waste at an off-site location. There are a total of five (5) drying beds requiring maintenance and sludge removal, four (4) located at the NAS I WWTP, Bldg. 229 and one (1) at the NAS II WWTP, Bldg. 402. The Contractor shall not remove or compact sand during sludge removal. Upon completion of the sludge removal, the beds shall be raked level. The Contractor shall submit a waste manifest for each dry bed sludge removal and disposal, per Section F. Informational Note: The depth of sludge to be removed ranges from five to ten centimeters and the quantity of sludge to be removed per occurrence is approximately ten cubic meters. The Government has found on prior contract that a frequency of three for each dry bed sludge removals and maintenance per year has provided an adequate service level (annually).	Sludge and vegetation is removed and disposed of per environmental regulations. Drying beds are fully operational after completion of services.

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4	Non-Recurring Work	Non-Recurring work may be ordered utilizing DoD FedMALL in accordance with Section H or on a task order in accordance with the PROCEDURES FOR ISSUING ORDERS clause in Section G. The order will specify the exact locations and types of work to be accomplished. The period of performance will be specified in each order.	Refer to Non-Recurring ELINs for task listings, descriptions and related requirements. All periods of performance are measured from issue date of order to acceptance of the work.	Performance Standards for Non-Recurring work will be the same as those in Spec Item 3 where applicable.